BACHELOR'S THESIS

Modelling Wellbeing: Comparing Statistical, Machine Learning, Hybrid and Bayesian Network Models on Psychological Data

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Chapter 1 - Introduction

1.1. Motivation

Mental health and psychological wellbeing are among the most complex and consequential outcomes in modern society. Conditions such as depression, chronic stress, and low life satisfaction affect hundreds of millions of people worldwide, with significant consequences for individuals, organisations, and public health systems. Despite a broad scientific consensus on the factors that contribute to wellbeing - including social support, economic security, physical health, and psychological resilience - the question of how accurately and interpretably these factors can be modelled using computational methods remains open.

Two parallel challenges motivate this thesis. The first is the distinction between individual-level and societal-level data. Survey responses from employees or students capture rich personal experiences but are characterised by high noise, subjective bias, and complex interactions among predictors. Aggregate indicators such as GDP per capita, life expectancy, and institutional trust capture structural conditions at the country level but abstract away individual variation. Whether predictive models perform differently across these two levels of analysis - and whether the same modeling frameworks can be applied consistently to both - is a question with both methodological and practical implications.

The second challenge concerns the tension between prediction and explanation. Machine learning models have demonstrated strong predictive performance across a wide range of psychological and health outcomes, but their internal logic is often opaque. Statistical models offer interpretability but may miss nonlinear relationships. Bayesian Networks occupy a distinctive position: they are generative probabilistic models that can encode causal structure explicitly, enabling reasoning under uncertainty in a way that neither regression nor ensemble methods can match. Whether Bayesian Networks can recover plausible causal structures from observational wellbeing data - and whether those structures add value beyond what machine learning feature importance already reveals - is the central methodological question of this thesis [Pearl2009, Scutari2014].

These challenges are not purely academic. Policymakers designing mental health interventions, employers evaluating workplace support programs, and researchers studying the determinants of happiness all need models that are not only accurate but interpretable and causally grounded. A model that predicts depression with high accuracy but cannot explain which factors drive the prediction, or in which direction, is of limited practical value for intervention design [Shmueli2010, Breiman2001].

1.2. Scope

This thesis presents a systematic comparative study of statistical, machine learning, hybrid, and Bayesian Network models for predicting and explaining psychological wellbeing under uncertainty, applied consistently across both individual-level and societal-level data.

The study is grounded in four datasets spanning two levels of analysis. At the individual level, three datasets are examined: a survey of mental health conditions in the technology industry, a survey of depression among students, and a Canadian population health survey measuring life satisfaction. At the societal level, the World Happiness Report Gallup Poll dataset - covering 149 countries across 2005–2020 - provides a macro-level comparison. An additional synthetic Kaggle dataset nominally derived from the World Happiness Report is included as a data quality case study, demonstrating the consequences of absent signal for all modelling approaches.

The scope is deliberately comparative. The same modeling frameworks are applied consistently across all datasets, enabling direct cross-dataset and cross-model evaluation. The study does not aim to perform clinical diagnosis or psychological assessment. Its focus is on modelling self-reported mental health and wellbeing outcomes using different predictive approaches, with particular emphasis on the causal structure of the data and the interpretability of model outputs.

Wellbeing is operationalised across multiple constructs depending on the dataset: work interference and treatment-seeking behaviour in the workplace context [Diener1984], depression as a binary clinical indicator in the student context, life satisfaction as a continuous self-report measure in the Canadian survey [Pavot1993], and the Cantril Life Ladder score as a societal happiness measure in the Gallup dataset [Helliwell2023, OECD2020].

1.3. Research Questions

The study is organised around five research questions that together address the comparative performance, interpretability, causal validity, and robustness of the modelling frameworks examined:

RQ1. How do different modeling approaches perform when predicting wellbeing at individual versus societal levels? This question examines whether predictive performance - measured by classification metrics at the individual level and regression metrics at the societal level - differs systematically between data levels, and whether that difference varies across model families.

RQ2. How does interpretability differ between regression, machine learning, and Bayesian Networks? This question examines the qualitative and quantitative differences in how each model family explains its predictions, using coefficient analysis, SHAP values, feature importance rankings, and Bayesian Network conditional probability tables as complementary interpretability tools [Lundberg2017, Molnar2019, Goldstein2015].

RQ3. Are Bayesian Networks better at capturing plausible causal structures in wellbeing data? This question evaluates whether Bayesian Network structure learning, with and without expert constraints, recovers directed acyclic graphs whose edge directions are consistent with established domain knowledge about the causal determinants of mental health and wellbeing [Pearl2009, Morgan2015].

RQ4. Can Bayesian Networks reveal causal structures that ML models cannot? This question examines whether the explicit directional edges learned by Bayesian Networks provide causal information beyond what machine learning feature importance conveys, and what the conditions are under which data-driven structure learning succeeds or fails [Yarkoni2017, Scutari2014].

RQ5. How robust are these models to missing data, noise, and correlated predictors? This question evaluates model stability across cross-validation folds, sensitivity to the removal of dominant predictors, and performance under conditions of high missingness and correlated feature sets [Hastie2009, Kuhn2013].

1.4. Overview of Approach

The methodology follows a consistent pipeline applied to all datasets. Each dataset is preprocessed in two versions: a machine learning version with imputation, one-hot encoding, and missingness indicators, and a Bayesian Network version preserving missing values as discrete states for structure learning. Baseline machine learning models - logistic or linear regression, decision tree, random forest, and XGBoost - are trained and evaluated using stratified cross-validation with multiple metrics. Hybrid models combining nonlinear feature selection with linear estimators, and stacking ensembles, are evaluated against the baselines. Bayesian Networks are learned using Hill-Climb Search with BIC scoring, both unconstrained and with expert-defined forbidden edges encoding domain knowledge about causal ordering.

Interpretability is analysed at two levels. At the global level, SHAP TreeExplainer values provide model-agnostic feature importance and directional effect estimates for the best-performing model on each dataset. Sensitivity analyses remove dominant proxy predictors to assess robustness. At the structural level, Bayesian Network graphs are inspected for causally implausible edges, and BIC scores are compared between unconstrained and constrained versions to quantify the statistical cost of expert constraints.

Cross-dataset comparison is performed through unified performance tables, visualisations of hybrid model gain over baselines, and a Bayesian Network consolidation summarising structure quality, inference results, and causal pathway robustness across all datasets. The thesis concludes with a discussion connecting empirical findings to the five research questions and to the broader prediction-explanation debate in psychological science [Breiman2001, Shmueli2010].

Chapter 2 – Literature Review

2.1. Overview

This chapter situates the thesis within three intersecting bodies of literature: the debate between predictive and explanatory modelling in statistical and psychological science, the theory and practice of Bayesian Network causal discovery, and the measurement of subjective wellbeing at individual and societal levels. Together these three strands provide the conceptual foundation for the comparative modelling framework developed in this thesis and motivate the five research questions introduced in Chapter 1.

2.2. Prediction vs Explanation Debate

The relationship between statistical explanation and predictive accuracy has been a central tension in quantitative science for decades. The distinction was formalised most sharply by Breiman [Breiman2001], who identified two cultures in statistical modelling: the data modelling culture, which assumes a stochastic data-generating process and prioritises interpretable parametric models, and the algorithmic modelling culture, which treats the process as unknown and prioritises predictive accuracy. Breiman argued that psychology and the social sciences had become overly committed to the data modelling culture at the expense of predictive validity, and that algorithmic methods such as ensemble trees and neural networks offered substantial gains in out-of-sample performance.

Shmueli [Shmueli2010] developed this distinction more formally by introducing a framework for distinguishing explanatory modelling - aimed at testing causal hypotheses about the data-generating process - from predictive modelling - aimed at producing accurate forecasts of new observations. She argued that the two goals require different model selection criteria, different evaluation metrics, and different interpretive standards, and that conflating them leads to both poor predictions and poor causal inference. A model that fits the training data well and has theoretically grounded coefficients may nevertheless perform poorly on held-out data, while a model with strong out-of-sample performance may be causally uninterpretable. This framework directly motivates the comparative design of this thesis, which evaluates both predictive performance and causal interpretability across model families.

Yarkoni and Westfall [Yarkoni2017] extended this argument specifically to psychological science, demonstrating that psychology's near-total focus on explaining behaviour through controlled experiments had produced a literature of intricate causal theories with little predictive validity. They proposed that adopting machine learning principles - in particular regularisation, cross-validation, and out-of-sample evaluation - would produce more accurate and generalisable psychological models. Their argument is directly relevant to the wellbeing prediction context: the variables that best explain depression or life satisfaction in a theoretical framework may not be the same as those that best predict it in a held-out sample.

The practical implication of this debate for the present thesis is that no single modelling approach is optimal across both goals simultaneously. Logistic and linear regression models offer coefficient-based explanations grounded in theory but may miss nonlinear relationships. Ensemble methods such as Random Forest and XGBoost offer strong predictive performance but require post-hoc interpretability tools. Bayesian Networks offer explicit causal structure but may sacrifice predictive accuracy in exchange for interpretability. Understanding this trade-off empirically - across multiple datasets and outcome types - is the central contribution of the present work.

The growing adoption of complex machine learning models in applied settings has generated a parallel literature on post-hoc interpretability methods. Hastie, Tibshirani, and Friedman [Hastie2009] provide the statistical foundation for tree-based and ensemble methods, including the conditions under which feature importance measures are reliable. Friedman [Friedman2001] introduced gradient boosting machines and partial dependence plots as tools for understanding the marginal effect of predictors in nonlinear models.

Goldstein and colleagues [Goldstein2015] introduced individual conditional expectation plots as a refinement of partial dependence plots, enabling visualisation of heterogeneous predictor effects across individuals rather than only average marginal effects. Lundberg and Lee [Lundberg2017] unified feature importance methods under the SHAP framework, providing a theoretically grounded decomposition of model predictions into additive feature contributions based on Shapley values from cooperative game theory. SHAP values satisfy desirable axiomatic properties - local accuracy, consistency, and missingness - that earlier importance measures do not, making them the standard tool for explaining individual predictions from tree-based models. Molnar [Molnar2019] provides a comprehensive overview of the interpretable machine learning landscape, situating

SHAP alongside partial dependence plots, permutation importance, and local surrogate models within a unified taxonomy of explanation methods.

In this thesis, SHAP is applied as the primary interpretability tool for the best-performing ML model on each dataset, complementing the structural interpretability provided by Bayesian Network graphs and coefficient tables from linear models.

2.3. Bayesian Networks Causal Discovery

Bayesian Networks are directed acyclic graphical models that represent a joint probability distribution over a set of variables through a product of conditional distributions [Scutari2014]. Each node represents a variable and each directed edge represents a conditional dependency, encoding the assertion that a child variable is directly influenced by its parent variables given the remaining network structure. The absence of an edge between two nodes encodes conditional independence, making Bayesian Networks a natural language for expressing causal hypotheses about data-generating processes.

Pearl [Pearl2009] established the theoretical foundations for causal inference in graphical models, introducing the do-calculus as a formal framework for reasoning about interventions - the effect of setting a variable to a fixed value - as distinct from observations. His framework distinguishes between associational quantities, which can be estimated from observational data, and causal quantities, which require either experimental manipulation or structural assumptions encoded in a directed acyclic graph. The d-separation criterion provides a graphical test for conditional independence, enabling the derivation of testable implications from a causal model without intervention data.

Morgan and Winship [Morgan2015] situate Bayesian Networks and graphical causal models within the broader counterfactual tradition of causal inference, connecting Pearl's do-calculus to the potential outcomes framework of Rubin and Neyman. They discuss the conditions under which observational data can support causal claims, emphasising the role of structural assumptions - encoded as forbidden or required edges in a Bayesian Network - in making identification possible.

Structure learning algorithms attempt to recover the graph structure of a Bayesian Network from observational data. Score-based methods such as Hill-Climb Search evaluate candidate graphs using a scoring function - typically the Bayesian Information Criterion - and search for the graph that maximises the score [Scutari2014]. The BIC penalises model complexity relative to sample size, favouring sparser graphs when data is limited. Constraint-based methods such as the PC algorithm use conditional independence tests to identify edges, while hybrid methods combine both approaches.

A fundamental limitation of data-driven structure learning is that it cannot distinguish between Markov equivalent graphs - directed acyclic graphs that encode the same conditional independence relationships but differ in edge direction. Without additional constraints derived from domain knowledge - such as the temporal ordering of variables or theoretical assumptions about causal direction - structure learning algorithms may recover statistically equivalent but causally implausible graphs. This limitation is directly relevant to the wellbeing context, where survey variables often measure co-occurring constructs whose causal ordering is theoretically motivated but not identifiable from observational data alone.

In this thesis, expert knowledge is incorporated into the structure learning process by specifying forbidden edges encoding known causal ordering constraints - for example, that demographic variables such as age and gender cannot be caused by psychological outcome variables, and that treatment-seeking behaviour cannot cause a family history of mental illness. The impact of these constraints on both statistical fit and causal plausibility is evaluated systematically across all datasets.

2.4. Wellbeing Measurement

Subjective wellbeing is a multidimensional construct encompassing cognitive evaluations of life quality, the presence of positive affect, and the absence of negative affect [Diener1984]. Diener's foundational review established subjective wellbeing as a legitimate scientific construct and identified its three primary components: life satisfaction as a cognitive judgement, positive affect as the frequency of pleasant emotional states, and negative affect as the frequency of unpleasant emotional states. This tripartite model has been the dominant framework in wellbeing research for four decades and provides the theoretical grounding for the outcome variables used in this thesis.

At the individual level, life satisfaction is most commonly measured using the Satisfaction With Life Scale [Pavot1993], a five-item instrument with established psychometric properties including high internal consistency, temporal stability, and convergent validity with related constructs. The review by Pavot and Diener confirms the scale's validity across diverse populations and its sensitivity to changes in life circumstances, making it appropriate for modelling as a continuous outcome variable. In the Canadian Survey dataset used in this thesis, life satisfaction is measured as a single-item numeric rating on a 0–10 scale, which is consistent with established single-item life satisfaction measures used in large-scale population surveys.

At the societal level, the World Happiness Report provides the most comprehensive annual assessment of national happiness across countries [Helliwell2023]. The primary measure is the Cantril Self-Anchoring Striving Scale - referred to as the Life Ladder - in which respondents rate their current life on a scale from 0 (worst possible life) to 10 (best possible life). The WHR framework identifies six explanatory factors that account for variation in national happiness scores: GDP per capita, social support, healthy life expectancy, freedom to make life choices, generosity, and perceptions of corruption. These factors are precisely the predictors available in the Gallup dataset analysed in this thesis, enabling a direct empirical test of whether their importance rankings recovered by machine learning and Bayesian Network models are consistent with the established WHR framework.

The OECD [OECD2020] provides a complementary framework for measuring societal wellbeing that extends beyond happiness to include material conditions, quality of life, sustainability, and inequality. The OECD guidelines distinguish between objective indicators - income, employment, housing - and subjective indicators - life satisfaction, affect balance, meaning - and emphasise the importance of measuring both dimensions for a complete picture of population wellbeing. This distinction motivates the inclusion of both objective socioeconomic variables (income, food security, employment) and subjective psychological states (mental health, stress, sense of belonging) as predictors in the Canadian Survey models.

Mental health as a specific dimension of individual wellbeing has received increasing attention as both a public health priority and a modelling target. Depression is among the most prevalent mental health conditions globally, affecting an estimated 280 million people according to the World Health Organisation, and is associated with substantial reductions in quality of life, work productivity, and social functioning. The student depression and mental health in tech datasets used in this thesis capture two distinct at-risk populations - students facing academic and financial pressure, and technology workers facing workplace-specific mental health stigma and support access barriers - providing complementary individual-level perspectives on mental health outcomes.

Chapter 3 – Methodology

3.1. Overview

This chapter describes the methodological framework used to address the five research questions set out in Chapter 1. The study adopts a comparative design in which four families of models - statistical regression, machine learning, hybrid, and Bayesian Networks - are applied to five datasets varying in scale, measurement level, and task type. Two datasets involve classification of psychological outcomes at the individual level; two involve regression of life satisfaction at the individual level; and one involves regression of subjective wellbeing at the societal (macro) level. A fifth dataset, synthetically generated, is retained as a data quality case study rather than a substantive modelling exercise. The consistent application of the same model families across all datasets enables direct comparison of predictive performance, interpretability, and structural validity.

Section 3.2 describes each dataset and the preprocessing decisions applied to it. Section 3.3 introduces the four model families and their configurations. Section 3.4 covers the evaluation framework, including metrics, cross-validation strategy, and the handling of the evaluation asymmetry between ML and BN pipelines. Section 3.5 gives a full account of the Bayesian Network structure learning approach, covering the algorithm, scoring function, expert knowledge constraints, parameter estimation, and inference procedure.

3.2. Dataset descriptions and preprocessing decisions

Five datasets were selected to cover a range of wellbeing constructs, measurement instruments, and analytical challenges. The distinction between micro-level (individual survey data) and macro-level (country-aggregated data) is treated as a primary axis of comparison throughout the thesis.

3.2.1 Mental Health in Tech Survey

The OSMI Mental Health in Tech Survey is a publicly available dataset collected via self-report from technology-sector workers. After removing observations with missing target values, the working sample comprised 995 respondents and 43 features. The prediction target was a three-class variable, *combined_target*, constructed by crossing the work interference item (*work_interfere*) with whether the respondent had sought treatment: Class 0 indicates no interference, Class 1 indicates interference without treatment, and Class 2 indicates interference with treatment. The class distribution was approximately 60.6%, 21.4%, and 18.0% respectively.

Preprocessing for the machine learning pipeline included the addition of binary missingness indicators for features with non-trivial missing rates, followed by median imputation for continuous variables and mode imputation for categoricals. Nominal features were one-hot encoded with reference categories dropped to prevent multicollinearity. For the Bayesian Network pipeline, missing values were preserved as a distinct state rather than imputed, ordinal encodings were retained, and high-cardinality geographic features (*Country* and *state*) were excluded, as their large number of discrete states would render conditional probability tables sparse and unstable.

3.2.2 Student Depression Dataset

This dataset contains survey responses from students on academic, social, and psychological variables. After filtering to confirmed student respondents - removing 0.1% of rows identified as non-students - the sample comprised 27,870 observations. The binary target, *Depression_yes*, indicated presence or absence of depression. Four features were dropped prior to modelling: *City*, which contained a mix of city names and degree abbreviations indicating data entry inconsistency; *Profession*, redundant after student filtering; and *Work Pressure* and *Job Satisfaction*, which are not applicable to a student population. The final feature set comprised 23 variables.

A notable characteristic of this dataset is the dominance of the suicidal ideation feature, which accounts for approximately 52–56% of intrinsic variable importance in tree-based models. Because suicidal ideation is a co-symptom of depression rather than an antecedent cause, its concentration of predictive signal raises questions about whether the models are learning genuine risk-factor structure or simply detecting a correlated symptom. This is examined in sensitivity analysis. For the Bayesian Network, age and CGPA were discretised into bins to satisfy the discrete variable requirement; continuous features in the ML pipeline were retained in their original form.

3.2.3 Canadian Community Health Survey

This large-scale national health survey comprises approximately 104,000 respondents. The continuous target, *Life_satisfaction*, is measured on a 0–10 scale. Columns with more than 60% missing values were dropped prior to modelling, removing features such as BMI for the 12–17 age group, fruit and vegetable consumption, and smoking status. The target missingness indicator was excluded from the feature set to prevent leakage. After preprocessing, approximately 58 features were retained.

Two variables - *Mental_health_state* and *Gen_health_state* - are self-rated health variables that function as proxies for life satisfaction: they correlate strongly with the target but arguably reflect a similar evaluative process rather than being independent causal inputs. Their contribution is examined in a dedicated sensitivity analysis. *Food_security* is coded ordinally from 0 (fully food-secure) to 3 (severely insecure). *BMI_18_above* is coded as 1 (underweight or normal) and 2 (overweight or obese).

3.2.4 WHR Synthetic Dataset

This dataset, presented on Kaggle as a World Happiness Report synthetic dataset, contains 4,000 observations across 10 countries and 20 years. The *Happiness_Score* column was dropped prior to modelling due to direct target leakage. Initial modelling revealed that all model families returned R^2 values near zero, and the maximum absolute correlation between any feature and the target was 0.049. The dataset contains no genuine predictive signal and is therefore not used as a substantive modelling exercise. It is retained as a case study in data quality detection, illustrating how a consistent modelling pipeline can identify a null-signal dataset before interpretability or causal analysis is attempted. The methodological implications of this finding are discussed in Section 4.4.1.

3.2.5 WHR Gallup Dataset

The Gallup World Poll data underlying the World Happiness Report was obtained via Kaggle and represents the most credible macro-level dataset in this study. The working sample comprised 1,949 observations across 149 countries and the period 2005–2020. The target variable, *Life_Ladder*, is the Cantril self-anchoring scale, which asks respondents to place their current life on a ladder from 0 (worst possible life) to 10 (best possible life). Seventeen features were retained after preprocessing, including log GDP per capita, social support, healthy life expectancy at birth, freedom to make life choices, generosity, perceptions of corruption, positive affect, negative affect, year, and binary missingness indicators for all columns with missing values. The maximum absolute feature-target correlation was 0.783, confirming genuine predictive signal. For the Bayesian Network, all continuous variables were quantile-discretised into five equal-frequency bins; year was binned into five-year periods to reduce the number of discrete states while preserving temporal ordering.

3.3. Model families and configurations

Four model families are applied consistently across all datasets. Each family represents a distinct position on the trade-off between predictive performance and interpretability, and their comparison is central to Research Questions 1, 2, and 3.

3.3.1 Statistical Baseline Models

The statistical baseline models serve as the interpretability anchor for the study. For classification tasks, logistic regression is used; for regression tasks, ordinary least squares linear regression is used. These models are selected not because they are expected to achieve the highest predictive accuracy, but because they are maximally transparent: every predictor's effect is directly readable from its estimated coefficient, including direction and magnitude. This transparency makes them the natural reference point for the interpretability comparisons developed in later chapters.

A single decision tree (classifier or regressor depending on task) is also included in the baseline set. Decision trees occupy an intermediate position: they are non-parametric and capable of capturing non-linear relationships and interactions, yet their structure can be visualised and traversed directly. Maximum depth is constrained via cross-validation to limit overfitting.

3.3.2 Machine Learning Models

Two ensemble methods form the primary machine learning models. Random Forest constructs a large ensemble of decision trees, each trained on a bootstrap sample of the training data with a random subset of features considered at each split. Predictions are aggregated by majority vote (classification) or averaging (regression). The randomisation introduced at both the sample and feature level reduces variance relative to any individual tree, and the ensemble's feature importance scores - based on the aggregate reduction in node impurity across all trees - provide a global measure of variable relevance.

XGBoost implements gradient boosting, constructing trees sequentially such that each new tree is fitted to the pseudo-residuals of the current ensemble under a differentiable loss function.

Regularisation is applied through a learning rate (shrinkage), subsampling of rows and columns, and L1/L2 penalties on leaf weights. XGBoost achieves the highest or near-highest predictive performance across all datasets in this study. Its predictions are used as the basis for the SHAP interpretability analysis described in Section 3.4.

Both models are evaluated using five-fold cross-validation with stratification on the target variable. Hyperparameters are held at well-established defaults throughout; the primary goal is comparison across model families rather than maximum optimisation of any individual model.

3.3.3 Hybrid Models

Hybrid models combine the feature selection capacity of ensemble methods with the interpretability of linear models. Two selection-then-estimation pipelines are evaluated: one using Random Forest importance scores to select features, followed by logistic or linear regression on the reduced feature set; and one using XGBoost importance scores in the same role. A third hybrid uses stacking, training a logistic or linear regression meta-learner on the out-of-fold predictions generated by Random Forest and XGBoost as base learners.

The inclusion of hybrid models reflects a practically motivated question: whether the interpretability of a linear model can be recovered without substantially sacrificing the predictive power of ensemble methods. The cross-dataset finding that hybrid models yield negligible gains over the best ensemble baseline is itself a substantive result, discussed in Chapter 5.

3.4. Evaluation framework

Evaluation metrics are chosen to match the task type and to reflect the priorities established in the research questions.

For classification tasks - MH Tech and Student Depression - the primary metric is macro-averaged F1 score, which computes precision and recall separately for each class before averaging, weighting all classes equally regardless of their frequency in the data. This is appropriate given the class imbalance present in both datasets, where a majority-class classifier would score misleadingly well on accuracy. Area under the ROC curve (AUC) is reported as a threshold-independent complement; for the three-class MH Tech target, AUC is computed using a one-vs-rest strategy.

For regression tasks - Canadian Survey, WHR Gallup, and WHR Synthetic - the primary metric is R^2, the proportion of variance in the target explained by the model. Root mean squared error (RMSE) is reported alongside R^2 to provide a metric expressed in the original scale of the target variable, which aids interpretability of effect magnitudes.

All machine learning models are evaluated using stratified five-fold cross-validation, and the reported metrics are the mean values across the five held-out folds. Cross-validated metrics are the primary basis for performance comparisons, as they provide an estimate of out-of-sample generalisation rather than in-sample fit. Bayesian Network predictions are computed on the full dataset because the pgmpy inference pipeline does not readily support fold-level re-estimation of structure and parameters. This evaluation asymmetry - CV metrics for ML models versus full-data metrics for BNs - is acknowledged throughout the results and discussion chapters. Where direct numerical comparisons are made between ML and BN performance, this asymmetry is explicitly noted, as BN full-data metrics are expected to be modestly optimistic relative to cross-validated equivalents.

3.5. BN structure learning approach

Bayesian Networks are probabilistic graphical models that represent the joint distribution of a set of random variables as a directed acyclic graph (DAG). Each node in the graph represents a variable; each directed edge from node A to node B encodes a direct conditional dependence of B on A, with B's distribution specified as a conditional probability table (CPT) over the states of its parents. The full joint distribution factorises as the product of these local conditional distributions, which makes inference computationally tractable and the learned structure directly interpretable as a set of conditional independence claims.

All Bayesian Networks in this study are implemented using pgmpy version 1.1.0.

3.5.1 Structure Learning Algorithm

Structure learning - the task of identifying the DAG from data - is performed using the Hill Climb Search (HCS) algorithm. HCS begins with an empty graph and iteratively proposes single-edge operations: addition, removal, or reversal of a directed edge. At each step, the operation yielding the greatest improvement in a scoring function is accepted; the algorithm terminates when no single-edge operation improves the score. HCS is a score-based greedy search and does not guarantee a globally optimal solution, but it is computationally practical for the variable counts encountered in this study and has been shown to recover plausible structures in comparable applied settings (Scutari and Denis, 2014).

To prevent overfitting through excessive parent sets, the maximum in-degree - the number of parents permitted for any single node - is constrained to three. This limit reduces the combinatorial search space, prevents the learning of highly specific conditional distributions that would not generalise, and produces structures that are more readily interpretable.

3.5.2 Scoring Function

The scoring function used throughout is BIC-d, the Bayesian Information Criterion adapted for discrete data. For a DAG G and dataset D of n observations, the BIC score is:

$BIC(G, D) = \log P(D \mid G, \hat{\theta}) - (k / 2) \cdot \log n$

where $\log P(D \mid G, \hat{\theta})$ is the log-likelihood of the data under the maximum likelihood parameter estimates $\hat{\theta}$, and k is the total number of free parameters across all CPTs in the graph. The penalty term $(k / 2) \cdot \log n$ discourages the addition of edges whose likelihood gain does not outweigh the cost of the additional parameters they introduce. HCS maximises this score, meaning structures with more edges are only preferred when they provide a sufficiently better fit to justify their complexity.

The pgmpy 1.1.0 built-in BIC scorer is incompatible with the BDeu parameter estimates used in this pipeline due to internal inconsistencies in how the library computes sufficient statistics for mixed prior and ML estimation. BIC scores are therefore computed using a custom implementation that directly evaluates the log-likelihood of the data under the learned CPTs and subtracts the standard BIC penalty. This implementation was validated against manually computed scores on small example graphs prior to use.

3.5.3 Expert Knowledge Constraints

An unconstrained Hill Climb Search will sometimes recover edges that are statistically supported by the data but domain-implausible - for example, an outcome variable appearing as a parent of a demographic variable, or a time-invariant characteristic being caused by a time-varying one. To quantify this problem, an unconstrained BN is estimated for each dataset and all recovered edges are evaluated against domain knowledge. Implausible edges are catalogued and the implausibility rate (number of implausible edges as a proportion of total edges) is reported.

A restricted BN is then estimated by encoding domain constraints as a forbidden edges list, passed to the HCS algorithm via pgmpy's ExpertKnowledge interface. Forbidden edges prevent the algorithm from ever placing a directed edge in a specified direction between a specified pair of nodes during the search. The categories of forbidden edges applied in this study include: demographic or background variables as targets of outcome or symptom variables; time-invariant variables as targets of time-varying ones; and any edge whose direction would imply a causal pathway that is biologically, temporally, or theoretically impossible given the survey design.

For the Canadian Survey, where the proxy status of *Mental_health_state* and *Gen_health_state* is a concern, an additional strongly restricted BN is estimated with those variables removed entirely from the structure search, producing a three-way comparison: unconstrained, restricted, and strongly restricted.

The BIC cost of applying constraints is defined as the percentage change in BIC score between the unconstrained and restricted models:

BIC cost (%) = ((BIC_restricted − BIC_unconstrained) / |BIC_unconstrained|) × 100

A positive value indicates the restricted model achieves a better (higher) BIC score than the unconstrained model - a result that occurs when the unconstrained search has overfit to noise via implausible edges. A negative value indicates a fitness cost: the constraints force the algorithm away from statistically preferred structures. Reporting this cost across datasets provides a quantitative characterisation of how much the unconstrained algorithm's output conflicts with domain knowledge on each dataset.

3.5.4 Parameter Estimation

Conditional probability tables are estimated using the BayesianEstimator with a BDeu (Bayesian Dirichlet equivalent uniform) prior. BDeu assigns a uniform prior over the parameters of each CPT, distributed equally across all parent configurations, with the total prior weight controlled by the equivalent sample size hyperparameter. An equivalent sample size of 10 is used throughout, which provides mild smoothing of sparse cells without substantially distorting CPTs that are well-supported by data. This is particularly important for the MH Tech and WHR Gallup datasets, where some combinations of parent states have few observations in the sample.

3.5.5 Inference and Prediction

Probabilistic inference is performed using Variable Elimination (VE). VE is an exact inference algorithm that computes the posterior distribution of a query variable given observed evidence by systematically eliminating non-query, non-evidence variables through marginalisation. For a query variable with evidence e, VE returns the full conditional distribution P(target | e), from which the most probable state (for classification) or the expected value computed from bin midpoints (for regression) is used as the point prediction.

For regression targets that have been quantile-discretised into bins, the expected value prediction is:

$$\hat{E}[\text{target} \mid e] = \Sigma\_i \text{ midpoint\_i} \cdot P(\text{target} = \text{bin\_i} \mid e)$$

where the sum is taken over all bins and midpoint_i is the midpoint of the original continuous range corresponding to that bin. This converts the BN's categorical posterior into a scalar prediction that can be evaluated using standard regression metrics.

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